A.U 2025-2026

TP1 - Java platform

Objectives:

- Become familiar with the syntax of the Java language.
- Write a first program in Java language.

Java Compiler: Checks the code written in the .java file to the .class code

JVM: Reads and interprets the .class files and executes the code.

Garbage Collector: Works in the background and checks for objects that are no longer used in the program to maximize memory usage.

Java Development Kit (JDK): Includes a set of libraries that make it easier to write programs. (For more information: JDK API documentation)

Java Runtime Environment (JRE): Includes components used when executing the code. The JRE is included in the JDK.

Configuring the Java Development Environment

VS code
https://www.youtube.com/watch?v=Y3Z-s6ai9EA
Netbeans
https://www.youtube.com/watch?v=qQpMHVh00QU

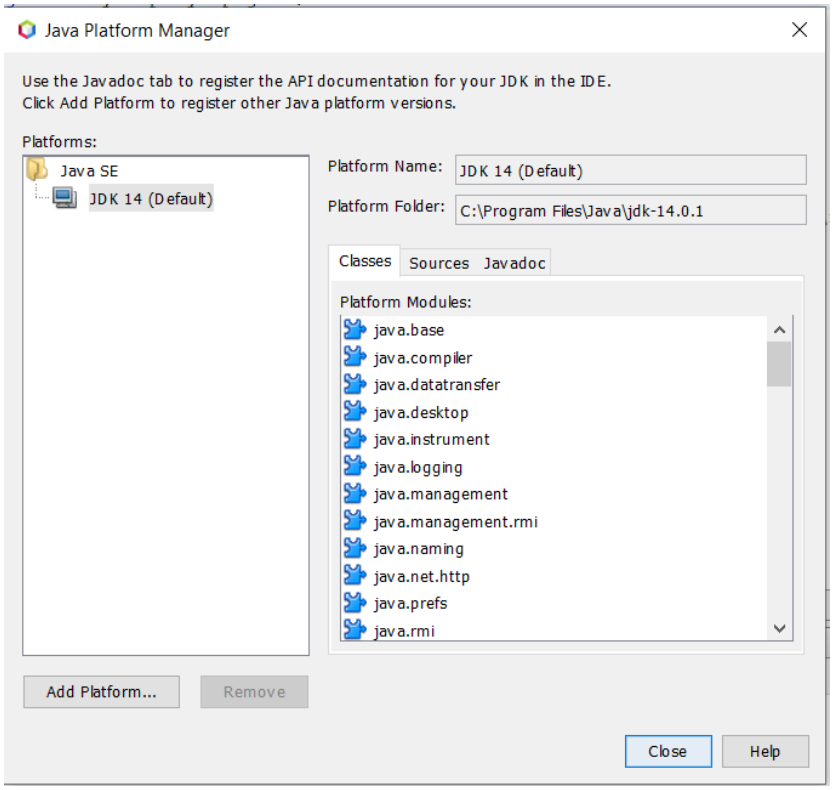
For Netbean

- Installer JDK: http://www.oracle.com/technetwork/java/javase/downloads/index.html
- Installer NetBeans

https://netbeans.apache.org/download/index.html?pagelang=fr

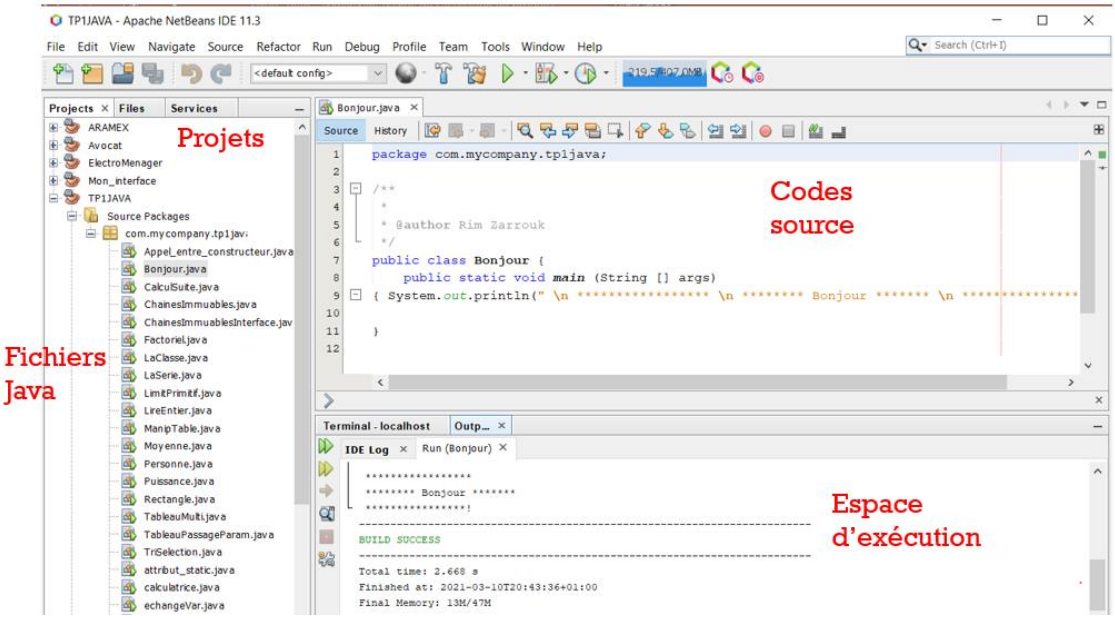
- Verification du JDK et JRE au niveau de IDE (Eclipse)

Tools > Java Plateforme



Les composants de NetBeans

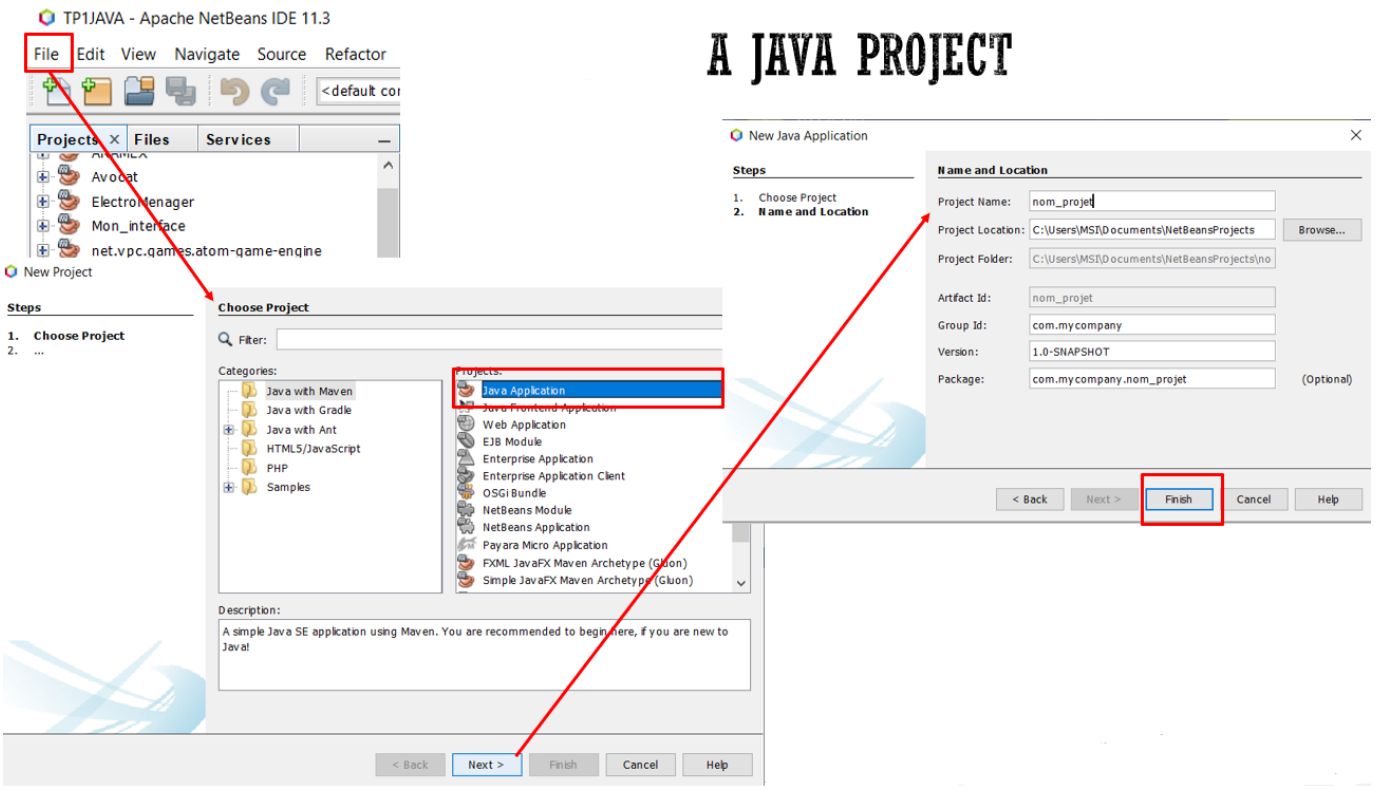
The NetBeans IDE has 4 components: Workspace, Projects, Perspectives and Views



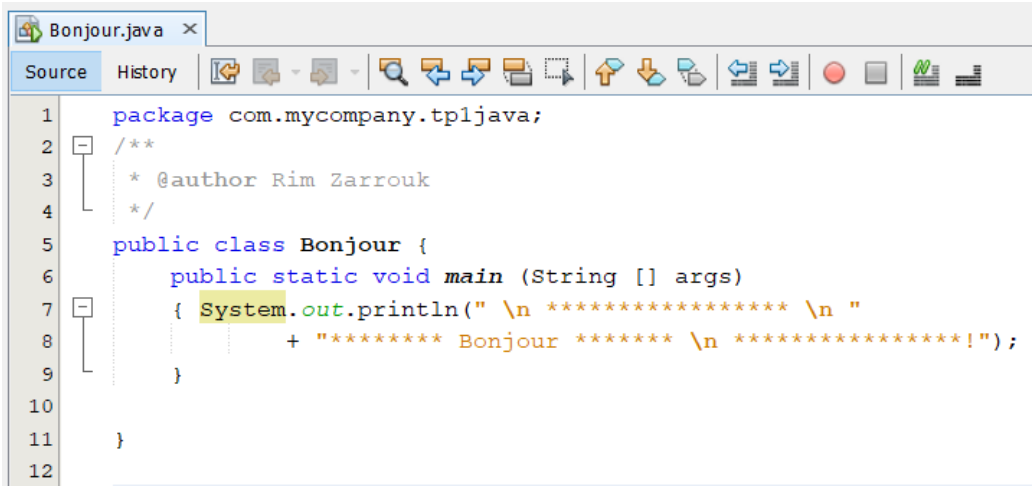
Presentation of a Java program

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A JAVA PROJECT



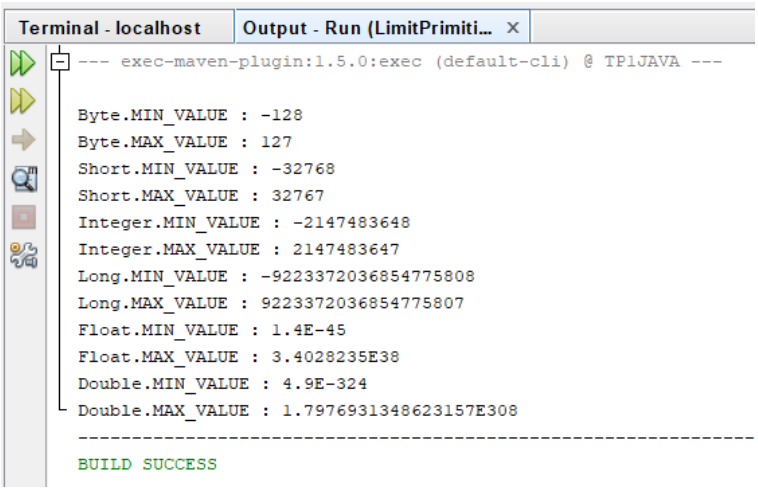
Structure of a Java program



Exercise 1:

Program for writing the limit values of primitive types:

Write the complete program giving the minimum and maximum limits of each of the integer and real types.



Exercise 2 :

A- Write a program that reads two variables a and b, then swaps their values and finally displays their new values.

B- Write a program that asks for the two dimensions of a rectangle, then calculates and displays its perimeter and area.

Exercise 3:

We will try to create a calculator application that allows to perform the following operations:

--Greet the user

--Ask for two integers from the standard input

--Perform subtraction, multiplication, addition and division operations using the two integers.

--Goodbye

Here is what you should get:

```
-------Good evening------
Please enter the first number: 5
Please enter the second number: 4
--------Calculate--------
5+4=9
5-4=1
4*5=20
5/4=1
-------Bye-------
```

Exercise 4 :

Write a program that reads two variables a and b, then swaps their values, and finally displays their new values.

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Homework: Number Guessing Game

Objective: Create a simple number guessing game where the program randomly generates a number between 1 and 100, and the user must guess it.

Instructions:

1. The program should generate a random number between 1 and 100.
2. The user is prompted to guess the number.
3. The program should give feedback if the guess is too low, too high, or correct.
4. The user should keep guessing until they guess the correct number.
5. After the correct guess, the program should print the number of attempts it took.